ABSTRACT

Neurosymbolic Verification of LLM-Generated Clinical Discharge and Treatment Plans

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Team, section and topic: Team 6, Section 22, Topic 12

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Language models now draft discharge and treatment plans, and the plans read well even when they are unsafe. General-purpose LLMs are already known to fail at formal planning tasks (Valmeekam et al., 2023); clinical stakes make that failure mode dangerous, not just inconvenient. A plan that discharges a patient before required labs return, or orders a contraindicated medication, still looks reasonable on the page. Care teams catch this by manual review, or by letting it run, which is too late. This project targets that gap with a checker that verifies LLM-generated care plans before a clinician or system acts on them. Intended users are clinical platform teams already piloting LLMs for discharge and care-plan drafting; scope for DATA 298A is a single hospital-encounter setting, not full clinical decision support.

We use MIMIC-IV and MIMIC-IV-Ext-CDM for admission, diagnosis, and treatment-sequence data. Both are public; access requires PhysioNet credentialing and CITI training, in progress for every team member.

The checker turns each plan into logic and tests it against rules the team writes, not the model, so a plan is never graded against constraints it invented. Four models will be built: Model A generates plans directly; Model B translates goals and clinical notes into formal specifications; Model C adds a solver that names the broken rule and returns it for repair; Model D pairs the LLM with a classical PDDL planner. Separately, the team reproduces the baseline in Hager et al. (2024), Figure 3 — 54.9% diagnostic accuracy for the best-performing open LLM on MIMIC-CDM's autonomous clinical decision-making task, across 2,400 patient cases — the comparison point, not one of the four models.

The primary metric is plan validity rate, checked against the rule set rather than a reviewer's read-through. Secondary metrics are constraint-violation escape rate, repair success rate, translation fidelity, and solver cost. Target: validity at or above 0.85 on 100 held-out cases, escape rate at or below 0.05.

By the end of DATA 298A the team will have the verifier, rule set, evaluation harness, benchmark, Model A implemented and evaluated, and the reproduced Hager et al. baseline, with Models B–D and the full four-model comparison completing in DATA 298B. The expected contribution is a domain-general safety layer that reduces unsafe plans reaching clinical review, with an honest account — via a secondary case study on software deployment pipelines — of which rules transfer and which do not.

Keywords: neurosymbolic verification; LLM planning; SMT solving; clinical discharge planning; counterexample-guided repair; formal specification

Project information

Item	Your entry
Abstract word count	397
Four proposed models	Model A (built in 298A): Direct LLM discharge/treatment plan generation. Model B (298B): Natural-language/clinical-note-to-formal-specification translation. Model C (298B): LLM generation with solver-in-the-loop counterexample repair. Model D (298B): Classical PDDL planner hybrid
Published baseline to reproduce	Hager et al., 2024 (Nature Medicine, peer-reviewed)
Exact baseline result location	Figure 3 — 54.9% diagnostic accuracy (best open LLM), MIMIC-CDM autonomous task, N=2,400 across 4 pathologies.
Benchmark or dataset	MIMIC-IV; MIMIC-IV-Ext-CDM (created by Hager et al., 2024 — the reproduced-baseline paper). Valmeekam et al. (2023) PlanBench cited as motivating evidence of general LLM planning failure, not as the reproduced baseline.
Dataset access status	Pending — PhysioNet credentialing (CITI training + Data Use Agreement) in progress for all members. Baseline reproduction code is public on GitHub (paulhager/MIMIC-Clinical-Decision-Making-Dataset/Framework/Analysis).
Primary evaluation metric(s)	Plan validity rate (primary). Secondary: constraint-violation escape rate, repair success rate, specification translation fidelity, solver overhead
Target success criterion	Plan validity ≥ 0.85 on a 100-case held-out split, with constraint-violation escape rate ≤ 0.05
GitHub repository URL	https://github.com/Akashkumarsenthil/Clinical-Discharge-Treatment-Plan-Verification
Linear board URL (instructor + ISA as admins)	https://linear.app/298a-team-6-neurosymbolic/team/298/overview

References

[1] de Moura, L., & Bjørner, N. (2008). Z3: An efficient SMT solver. In C. R. Ramakrishnan & J. Rehof (Eds.), Tools and Algorithms for the Construction and Analysis of Systems (pp. 337–340). Springer. https://doi.org/10.1007/978-3-540-78800-3_24

[2] Johnson, A. E. W., Bulgarelli, L., Shen, L., Gayles, A., Shammout, A., Horng, S., Pollard, T. J., Hao, S., Moody, B., Gow, B., Lehman, L. H., Celi, L. A., & Mark, R. G. (2023). MIMIC-IV, a freely accessible electronic health record dataset. Scientific Data, 10, 1. https://doi.org/10.1038/s41597-022-01899-x

[3] Valmeekam, K., Marquez, M., Sreedharan, S., & Kambhampati, S. (2023). PlanBench: An extensible benchmark for evaluating large language models on planning and reasoning about change. Advances in Neural Information Processing Systems, 36. Cited as motivating evidence of general LLM planning failure; not the reproduced baseline. https://arxiv.org/abs/2206.10498

[4] Hager, P., Jungmann, F., Holland, R., Bhagat, K., Hubrecht, I., Knauer, M., Vielhauer, J., Makowski, M., Braren, R., Kaissis, G., & Rueckert, D. (2024). Evaluation and mitigation of the limitations of large language models in clinical decision-making. Nature Medicine, 30, 2613–2622. https://doi.org/10.1038/s41591-024-03097-1